

ZETOD : TECH STACK WRITEUP

Overview

Every tool in ZeToD's stack was chosen based on three non-negotiable criteria — it must be free, open source, and production capable. No paid tiers, no trials, no hidden costs.

Frontend : React + Tailwind CSS

Hosted on Vercel (free tier)

React is the industry standard for building dynamic user interfaces and has the largest developer ecosystem of any frontend framework. This matters for an open source project because contributors are more likely to already know it. Tailwind CSS allows rapid UI development without writing custom CSS files, keeping the codebase clean and consistent. Vercel offers completely free hosting for open source projects with automatic deployments triggered by every GitHub push. Zero configuration required.

Backend : Node.js + Express

Hosted on Render (free tier)

Node.js was chosen because it uses JavaScript, the same language ecosystem as the frontend, reducing the context switching cost for developers contributing to both layers. Express is lightweight, unopinionated, and well documented making it ideal for building a clean REST API.

Render provides free backend hosting with automatic deploys from GitHub. It handles environment variables, HTTPS, and scaling without any configuration cost.

Database : PostgreSQL

Hosted on Supabase (free tier)

PostgreSQL is the most powerful open source relational database available. It supports full text search natively, which powers the question search and filtering features, without needing a separate search service. Supabase provides a free hosted PostgreSQL instance with a built-in dashboard for managing data, making it accessible to all contributors regardless of database experience.

Authentication : Supabase Auth

Hosted on Supabase (free tier)

Supabase Auth provides complete authentication infrastructure out of the box, sign up, login, password reset, JWT token management, and session handling. Building this from scratch would take weeks and introduce security risks. Using Supabase Auth means the team focuses on building the actual assessment engine instead of reinventing authentication. It is included in the same free Supabase instance as the database, no additional cost or service.

Code Execution : Judge0

Self hosted (completely free)

Judge0 is an open source code execution engine used by platforms like LeetCode and Codeforces. It runs user-submitted Python code in a secure sandbox environment with timeout controls, memory limits, and output validation, preventing malicious code from affecting the server.

Self hosting Judge0 means zero API costs regardless of how many assessments are run. It supports Python natively and returns stdout output for automated comparison against expected results.

Version Control : GitHub

github.com (free)

GitHub hosts the entire codebase publicly as an open source repository. It handles CI/CD pipelines through GitHub Actions, automatically running tests and triggering deployments to Vercel and Render on every push to the main branch. Being on GitHub also makes the project discoverable and contribution friendly.